

Musaad Alruwaili

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🌐 musaadalruwaili
📍 Alquryyat, Aljouf, Saudi Arabia

Professional Summary

- Ph.D.-qualified Electrical Engineer with over 8 years of experience spanning control systems, telecommunications, and industrial project management. Currently serving as a Project Engineer at AECC, with a career foundation in critical infrastructure at Saudi Electricity Company and Fluor Arabia.
- Expertise integrates advanced research in 5G/6G Device-to-Device (D2D) communication and security with practical industrial automation and SCADA integration. Proven ability to apply machine learning and optimization algorithms to complex engineering systems.

Employment History

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| Nov 2025 – Present | ■ Project Engineer, Advance Energy Company (AECC) , Turaif, Saudi Arabia. |
| Jun 2015 – Mar 2018 | ■ Control Systems Engineer, Fluor Arabia Limited Company , Al Khobar & Turaif, Saudi Arabia. |
| Sep 2006 – Sep 2007 | ■ SCADA Technician, Saudi Electricity Company , Al Jouf, Saudi Arabia. |
| Mar 2005 – Sep 2006 | ■ Communication Technician, Aircraft Accessories & Components Co. (AACC) , Tabuk, Saudi Arabia. |

Education

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| 2019 – 2025 | ■ Ph.D. in Electrical Engineering , The University of Toledo, Toledo, OH, USA. |
| 2013 – 2014 | ■ M.S. in Electrical Engineering , Gannon University, Erie, PA, USA. |
| 2009 – 2012 | ■ B.S. in Electrical Engineering , Gannon University, Erie, PA, USA. |

Technical Skills

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| Programming & Tools | ■ Python, MATLAB, Simulink, LabVIEW, PSpice, Git, Bash, L ^A T _E X |
| Control Systems & Automation | ■ State-space modeling, PID, LQR/LQG, frequency response, SCADA integration, industrial process optimization |
| Telecommunications | ■ 5G/6G, Device-to-Device (D2D) communication, wireless resource allocation, IoT systems |
| Artificial Intelligence | ■ Machine learning for engineering systems, predictive modeling, optimization algorithms |

Research Publications

1. **Alruwaili, M.**, Kim, J., & Oluoch, J. (2024). *Optimizing 5G Power Allocation With Device-to-Device Communication: A Gale–Shapley Algorithm Approach*. IEEE Access.
2. **Alruwaili, M.**, Kim, J., & Oluoch, J. (2025). *Enhancing Security and Privacy in 5G Device-to-Device Communication: A Secure Gale–Shapley Algorithm Approach*. IEEE Access.

Awards, Memberships & Certifications

Awards

- 2011  **Dean's List Scholar**, Gannon University.
- 2010  **President, Saudi Student Association**, Gannon University.

Professional Membership

- 2012  **IEEE Eta Kappa Nu (HKN)**, International Electrical Engineering Honor Society.

Certifications & Training

- 2015  Control Systems & Telecom Engineering Review, Fluor.
-  Control Systems Theory, Fluor.
-  Flow & Pressure Instrumentation, Fluor.
- 2016  Annual Ethics Certification, Fluor.

References

Available upon request.